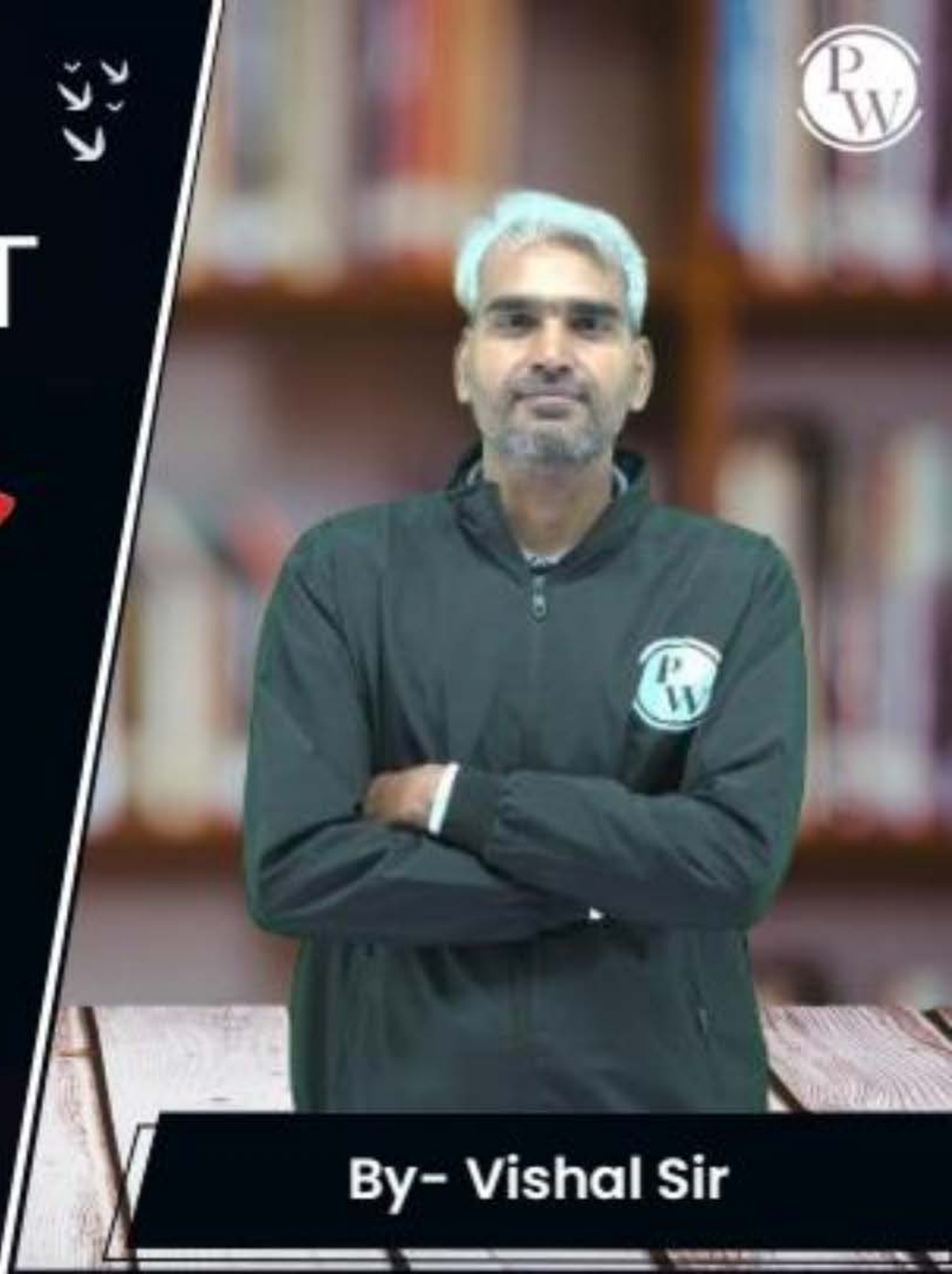
Computer Science & IT



Database Management System

ER Model

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- **Topic** ER model & ER diagram
- **Topic** Terminologies in ER diagram

Slide

Topics to be Covered



- Topic: ER model & ER diagram
- Topic: Integrity constraints

Slide



Topic : Mapping cardinality

- Mapping cardinality or cardinality ratio is used to denote the number of entities to which another entity can be associated through a certain relation set.

- For a binary relationship set mapping cardinalities must be one of the following types:
  1. One-to-one (1 : 1)
  2. One-to-many (1 : N)
  3. Many-to-one (N : 1)
  4. Many-to many (N : M)



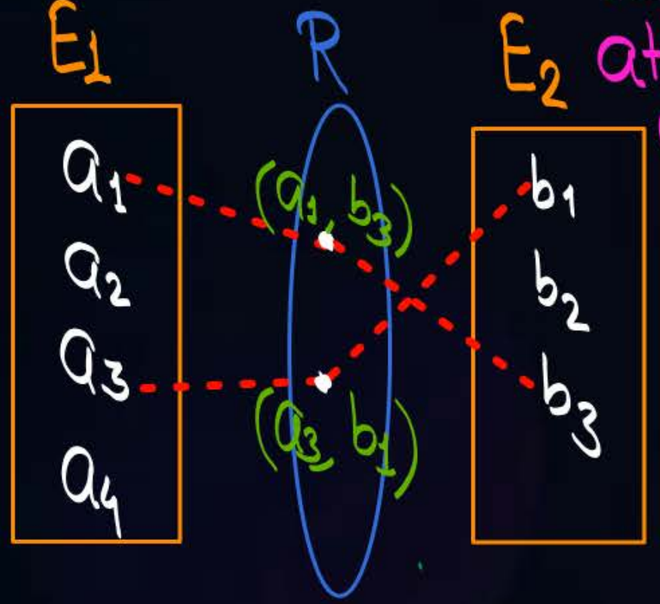
Topic : One-to-one

$E_1 : E_2 \equiv 1 : 1$



An Entity of set E_2 Can associate with at most one entity of set E_1

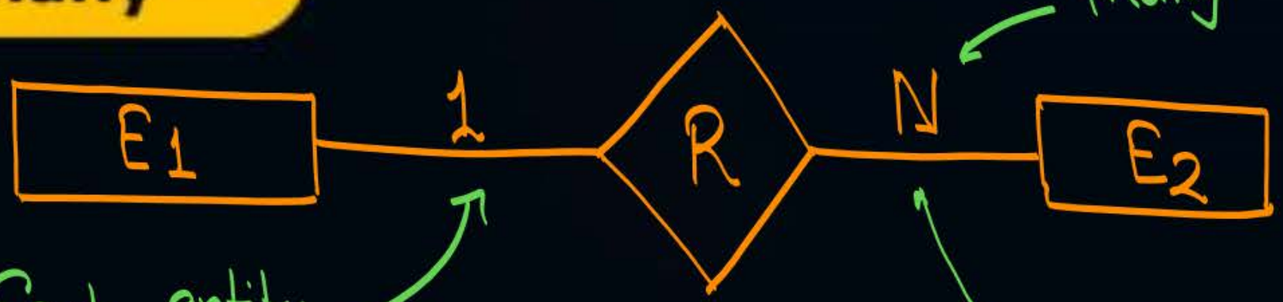
An entity of set E_1 Can associate with at most one entity of set E_2





Topic : One-to-many

$E_1 : E_2 = 1 : N$



Each entity of set E_2 can associate with atmost one entity of set E_1

↓

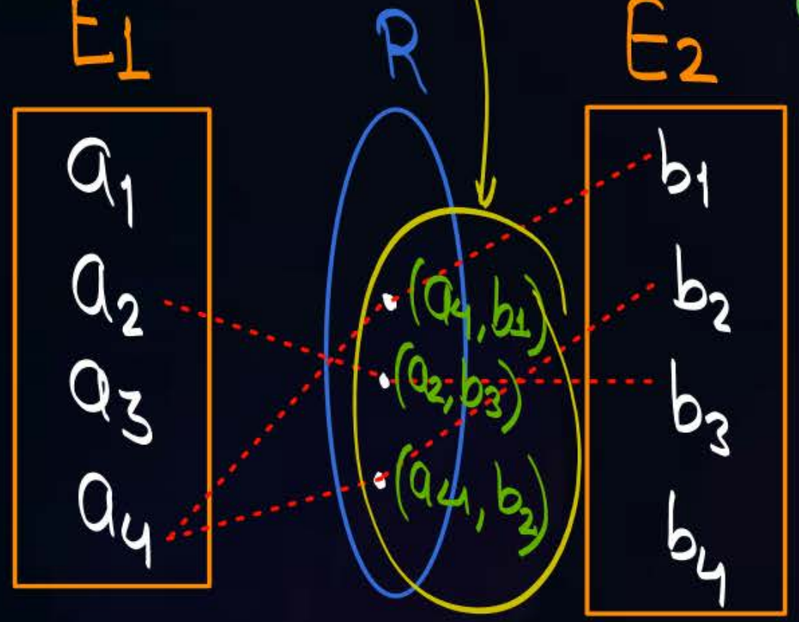
{ An Entity of set E_2 can appear at most once in relationship set }

Each entity of set E_1 can associate with at most N {i.e. 0 to N} {0 or more} entities of set E_2

↓

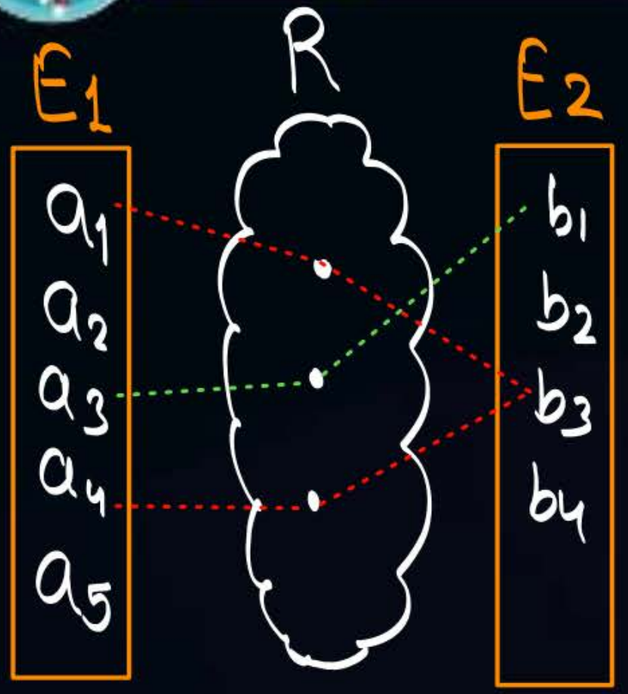
{ An Entity of set E_1 can appear any number of times in relationship set }

Relationship set





Topic : Many-to-one



$E_1 : E_2 \equiv N : 1$

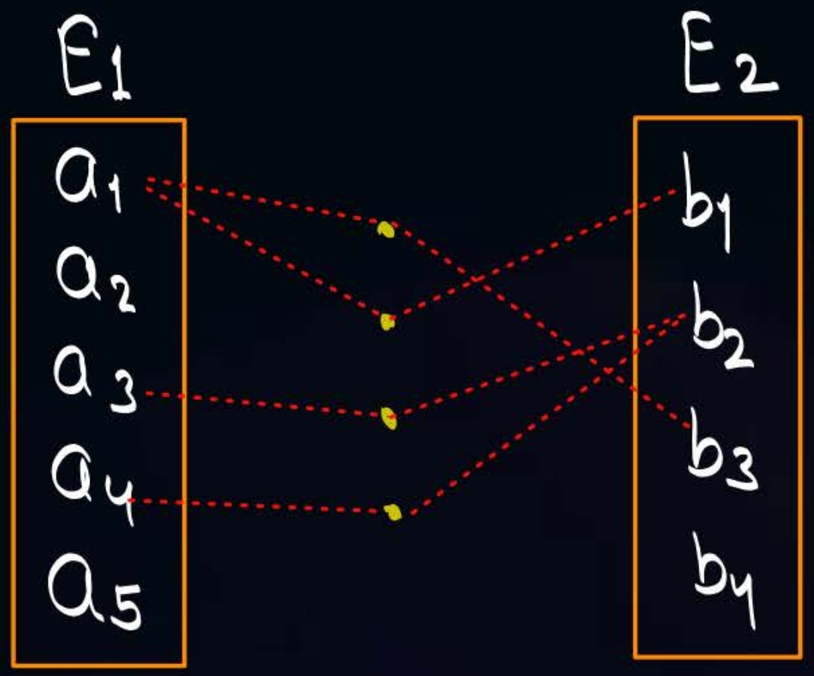


Each entity of set E_2 can associate with 0 or More (N) entities of set E_1

Each entity of set E_1 can associate with at most one entity of set E_2



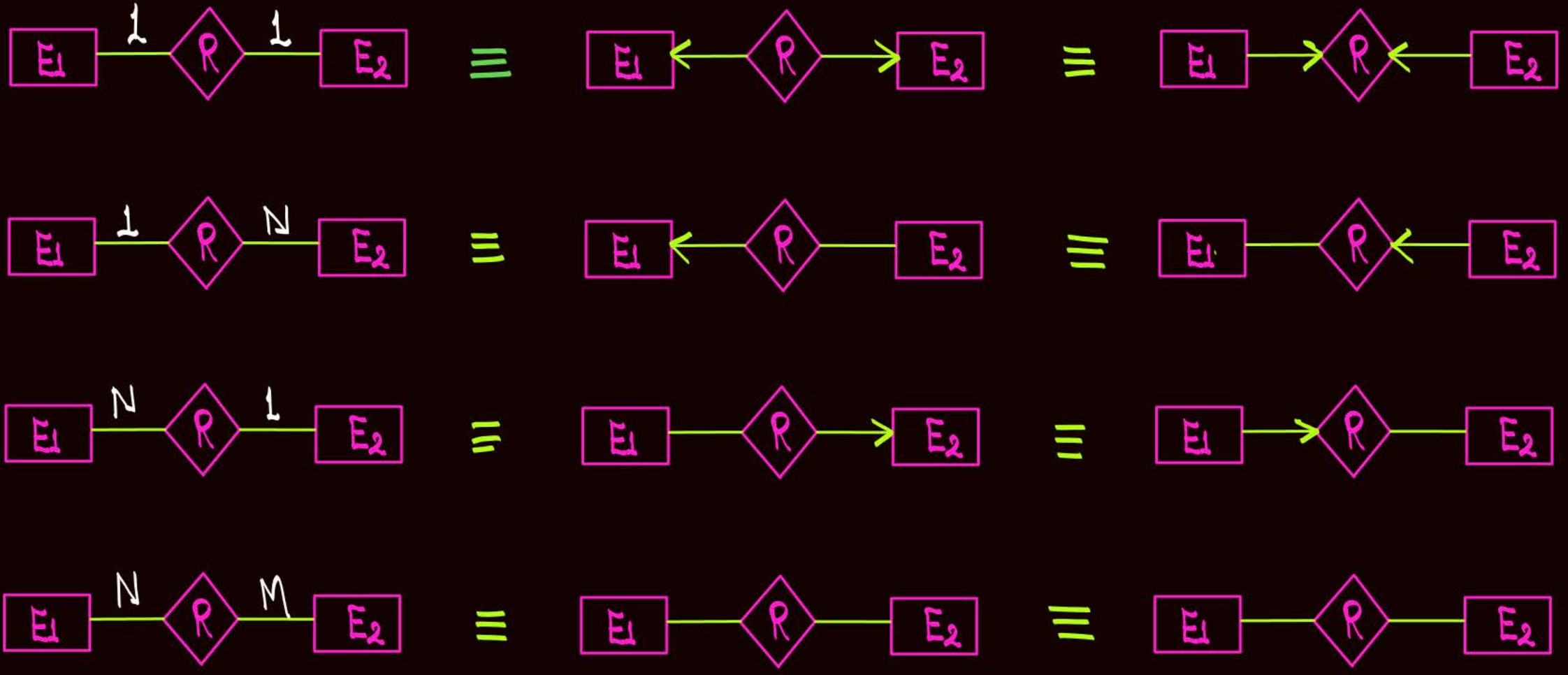
Topic : Many-to-many





Each entity of set E_2 can associate with 0 or more entities of set E_1

Each entity of set E_1 can associate with 0 or more entities of set E_2


E1
1
R
1
E2
≡
E1
R
E2
≡
E1
R
E2
E1
1
R
N
E2
≡
E1
R
E2
≡
E1
R
E2
E1
N
R
1
E2
≡
E1
R
E2
≡
E1
R
E2
E1
N
R
M
E2
≡
E1
R
E2
≡
E1
R
E2

* Min-Max Representation

Participation is represented using order pair of type (min, max)

1st value will be minimum value

2nd value will be maximum value

- Minimum number of times an entity can appear in the relationship set is represented by "min"

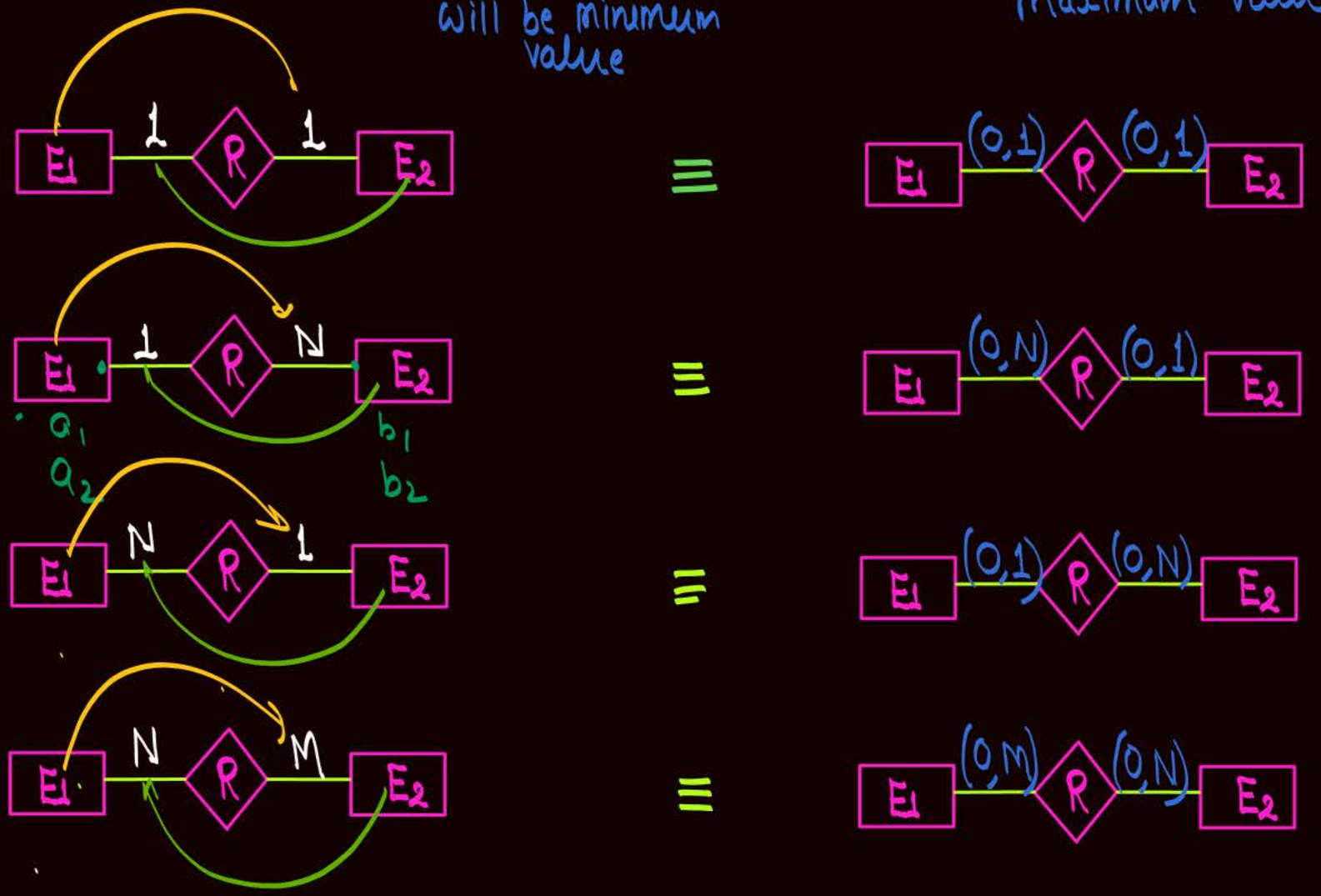
and " Maximum number of times an entity can appear in the relationship set is represented by "max"

Min-Max Representation

Participation is represented using order pair of type (min, max)

1st value will be minimum value

2nd value will be maximum value

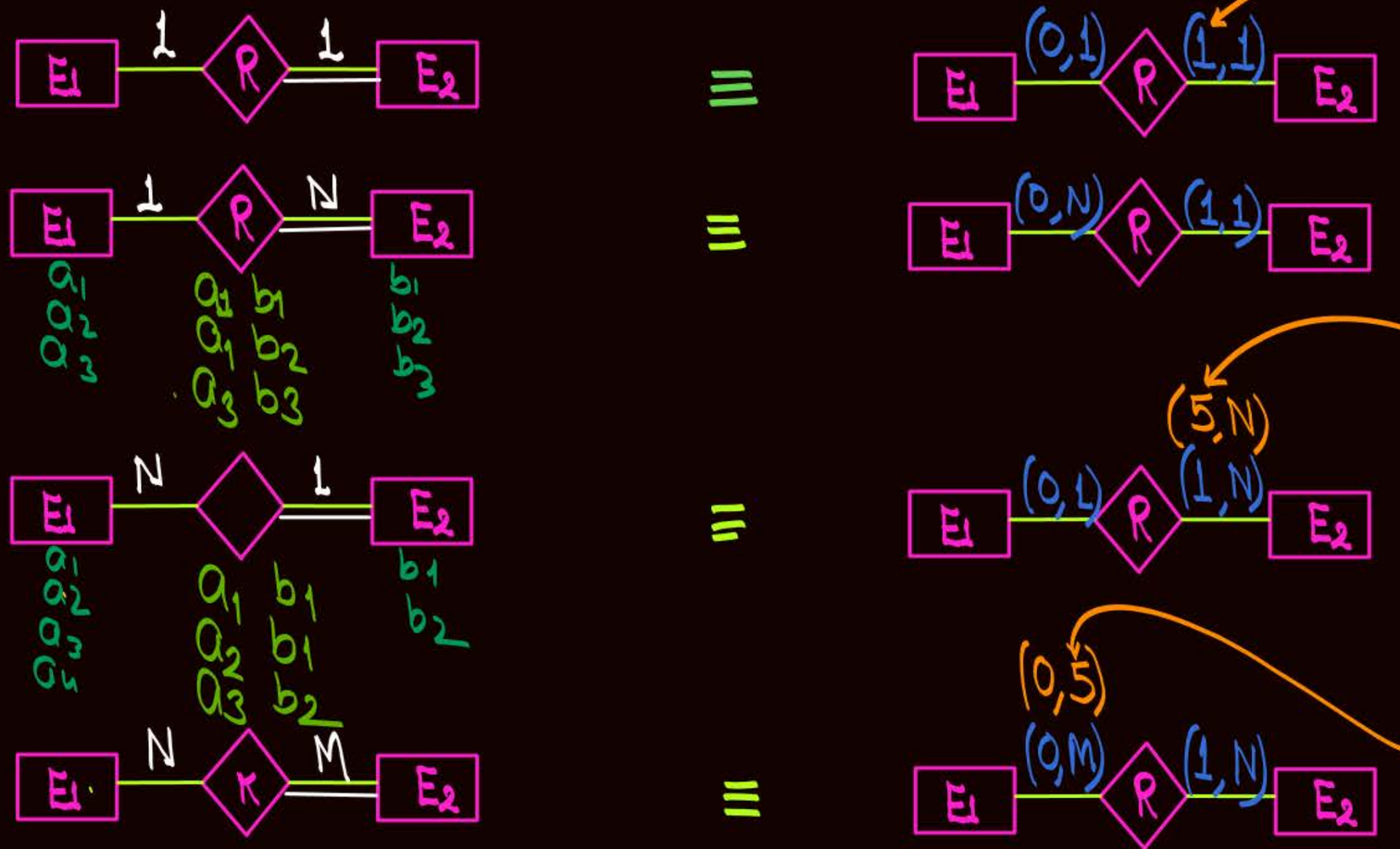


Min-Max Representation

Participation is represented using order pair of type (min, max)

1st value will be minimum value

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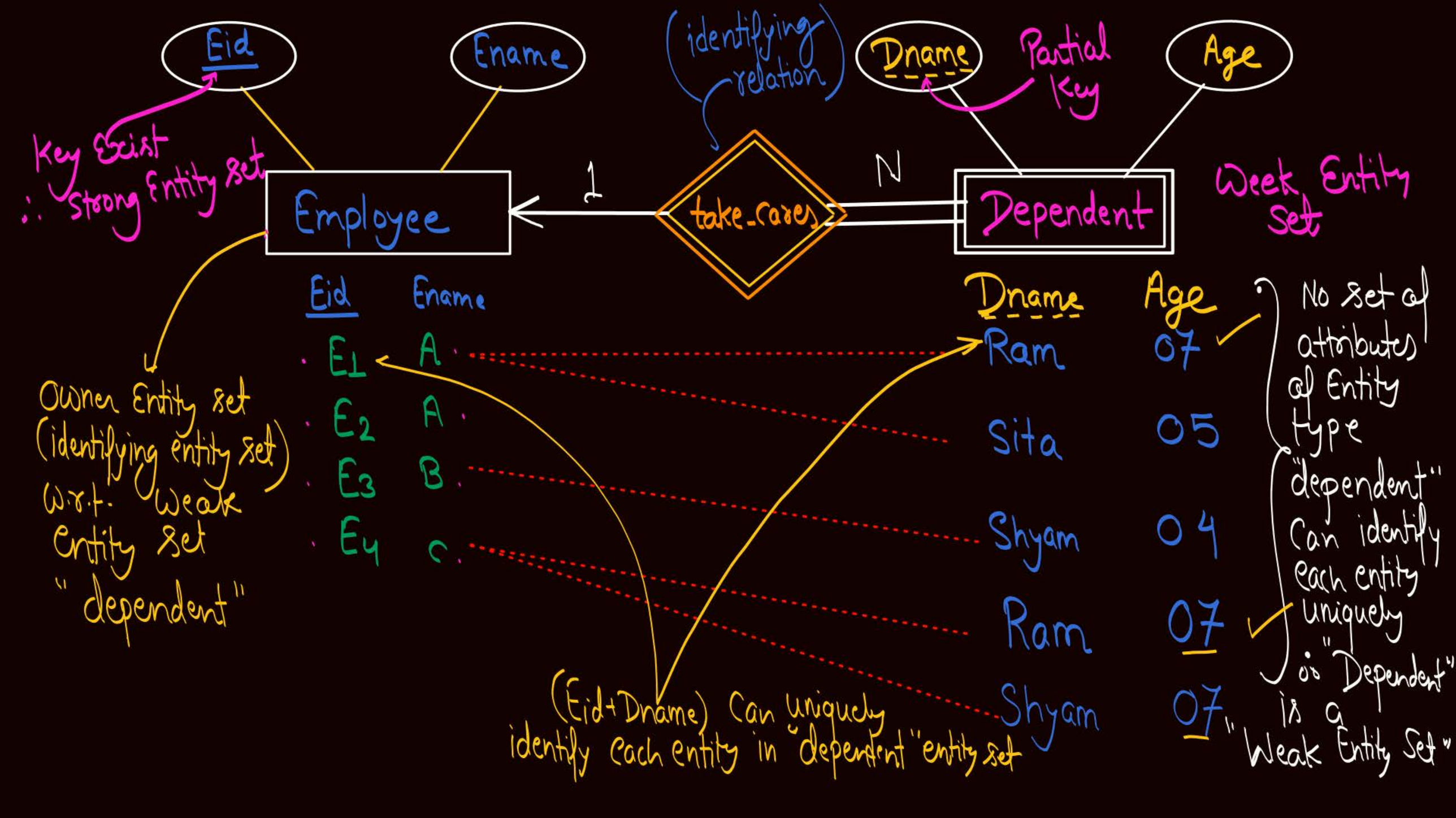


Topic : Strong entity Set

- A strong entity set is an entity set that contains sufficient attributes to uniquely identify all its entities. In other words, a primary key exists for a strong entity set.
- Primary key of a strong entity set is represented by underlining it.

Weak Entity Set :- An Entity set that does not Contain sufficient attributes to uniquely identify all its entities. then it is called Weak Entity set.

i.e. Primary key can not be defined


Eid
Ename
(identifying relation)
Dname
Partial key
Age
Key Exist
∴ Strong Entity Set
Employee
1
take_cares
N
Dependent
Week Entity Set
Eid
Ename
E1
A
E2
A
E3
B
E4
C
Dname
Age
Ram
07
Sita
05
Shyam
04
Ram
07
Shyam
07
Owner Entity set
(identifying entity set)
w.r.t. Weak
Entity set
"dependent"
No set of attributes of Entity type "dependent" Can identify each entity uniquely
∴ "Dependent" is a "Weak Entity Set"
(Eid+Dname) Can uniquely identify each entity in "dependent" entity set



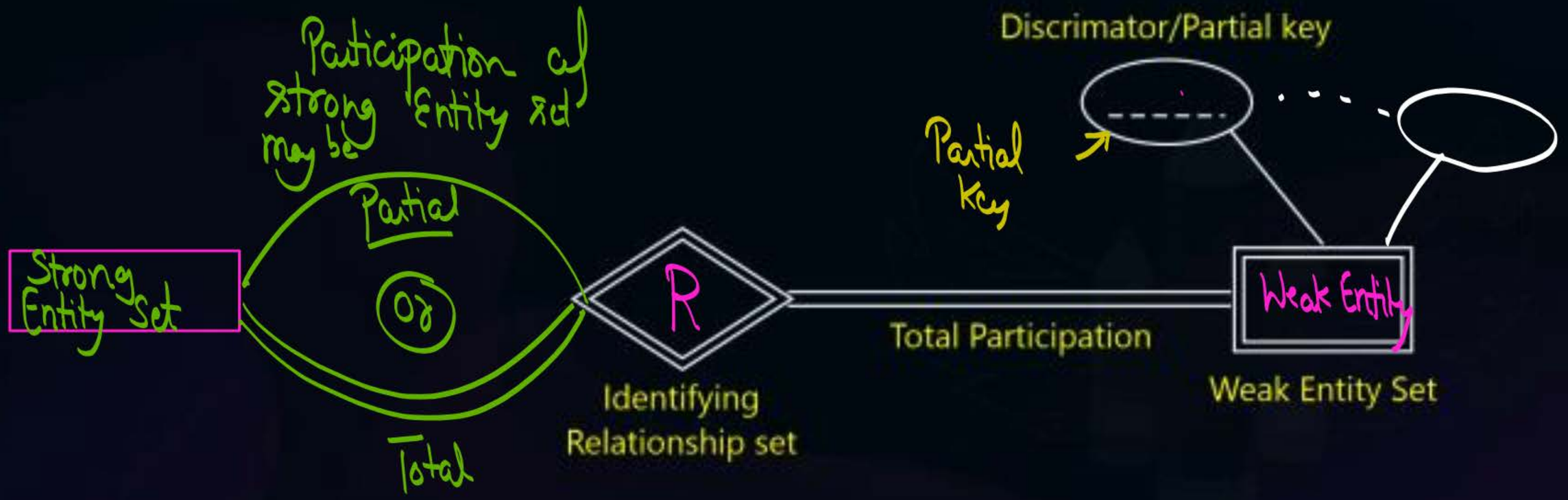
Topic : Differences between Strong entity set and Weak entity set

Strong entity set	Weak entity set
A single line rectangle is used for the representation of a strong entity set.	A double line rectangle is used for the representation of a weak entity set.
It contains sufficient attributes to form its primary key.	It does not contain sufficient attributes to form its primary key.
A single line diamond symbol is used for the representation of the relationship that exists between the two strong entity sets.	A double line diamond symbol is used for the representation of the identifying relationship that exists between the strong and weak entity set.
Total participation may or may not exist in the relationship.	Total participation always exists in the identifying relationship.



Topic : Weak entity set in ER diagram

- In general, mapping cardinalities b/w Strong and Weak Entity set will be "one to many"





Topic : Relational Model concept

- The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- The relational model is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Constraints

- These are the restrictions or set of rules imposed on the database content. It validates the quality of the database. It validates that various operations like data insertion, updation, etc., can be performed without affecting the integrity of the data.

- Constraints could be either on a column level or a table level. The column level constraints are applied only to one column, whereas the table level constraints are applied to the whole table.



Topic : Types of Constraints on Relational Model

1. Domain constraint
2. Key constraint or Tuple Uniqueness constraint
3. Entity Integrity constraint
4. Referential Integrity constraint



Topic : Domain constraint

(attribute)

- Every domain must contain atomic values(smallest indivisible units) which means composite and multi-valued attributes are not allowed.

- We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain. Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype



Topic : Key constraint or Tuple Uniqueness constraint

tuple

- These are called uniqueness constraints since it ensures that every tuple in the relation should be unique.

{ i.e., we must be able to define candidate key for the database }

A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys.



Topic : Entity Integrity constraint

- Entity integrity constraint specifies that no attribute of primary key must contain a null value in any relation.



Topic : Referential Integrity constraint

M. Imp

The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.

Some times may be on the same table



Topic : Referential Integrity constraint

- Foreign Key: A foreign key is a set of attributes in a table that refers to the primary key or alternate key of same table or some other table.

eg:

it references to the "Stu-id" attribute of Student relation

Foreign key

Primary key of student

Names of the attributes need not be same

Sid	Cid
S_1	C_1
S_1	C_2
S_3	C_2
S_4	C_3

Enroll

it is also a foreign key which will reference P.k. "Cid" of the course relation

(Relationship)

Enroll is the referencing relation

Stu-id	Sname	Branch
S_1		
S_2		
S_3		
S_4		

Student

{ Student is a referenced relation }

Cid	Cname	Branch
C_1		
C_2		
C_3		
C_4		

Course

Note:

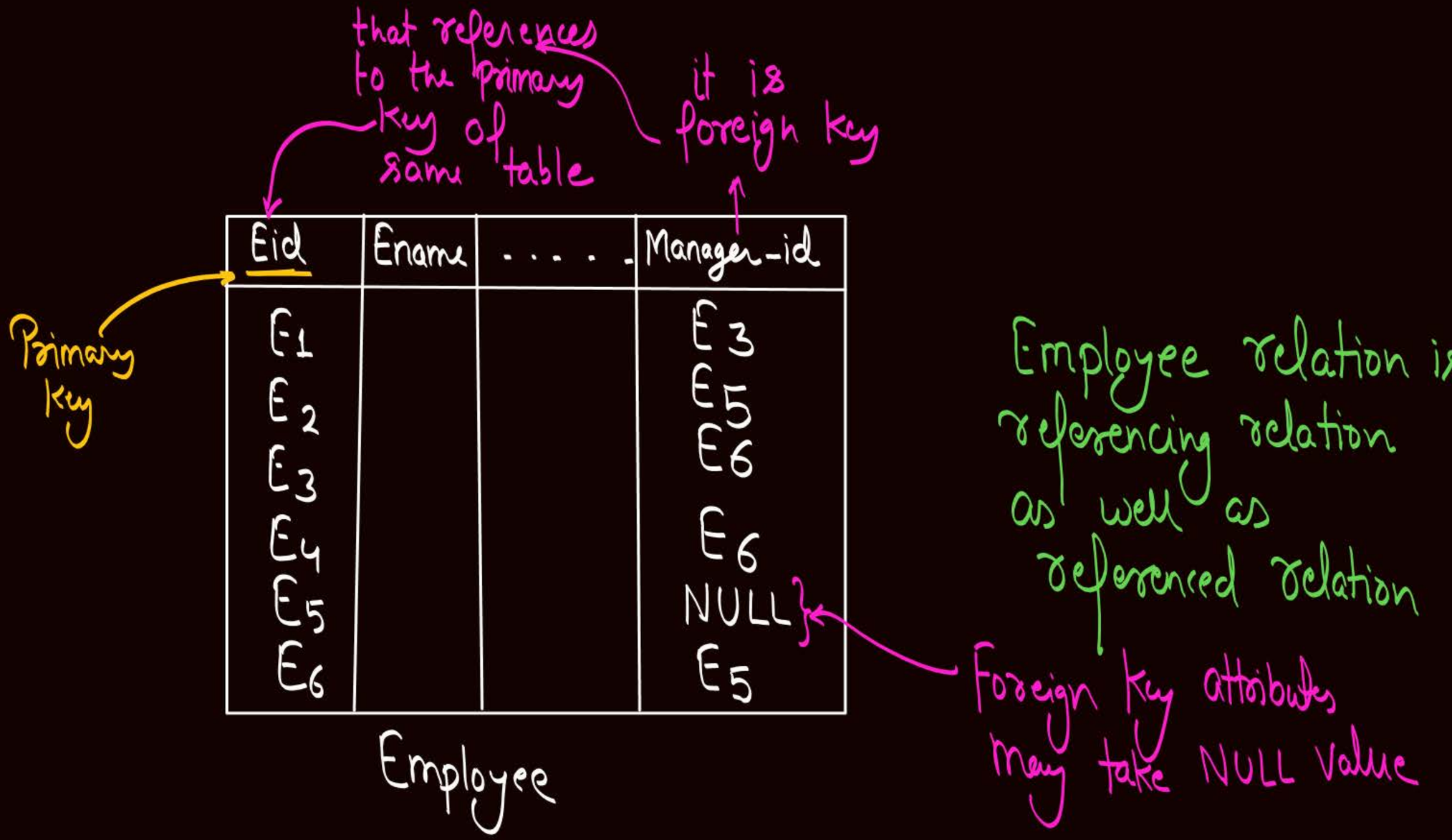
① A relation that contains foreign key is called a referencing relation

② Relation that contains the primary key / alternate key which is being referenced by the foreign key is called a referenced relation

In some cases,

referencing relation & referenced relation may be same

eg:


that references to the primary key of same table
it is foreign key
Primary key
Eid
Ename
. . . . .
Manager_id
E1
E2
E3
E4
E5
E6
E3
E5
E6
E6
NULL
E5
Employee
Employee relation is referencing relation as well as referenced relation
Foreign key attributes may take NULL value



Topic : Referential Integrity constraint

- Referential integrity constraint is enforced through a foreign key

∴ R_1 is referencing ∴ R_2 is Referenced.

- Let foreign key in relation R1 refers to primary key of relation R2.

The values of the foreign key in a tuple of relation R1 can either take the values from the primary key (values of) some tuple in relation R2, or can take NULL values, but can't be empty.

Values in foreign key Column $\subseteq$ [Values in Primary key/Alternate key column of referenced relation $\cup$ {NULL}]



2 mins Summary

Topic ER model & ER diagram

Topic Integrity constraints

Slide

THANK - YOU